

```

transmissivity.py x Henyey-Greenstein.py x
1 import numpy as np
2 from scipy.optimize import curve_fit
3 import matplotlib.pyplot as plt
4
5 def rayleigh(cos_theta): 1用法
6     return (3/4)*(1+cos_theta**2)
7
8 def HG(cos_theta,g):
9     return (1-g**2)/((1+g**2-2*g*cos_theta)**(3/2))
10
11 def TTHG(cos_theta,g): 2用法
12     return (HG(cos_theta,g)+HG(cos_theta,-g))/2
13
14 theta = np.linspace(start=0,2*np.pi,num=100)
15 cos_theta = np.array([np.cos(th) for th in theta])
16 P_rayleigh = rayleigh(cos_theta)
17
18 g0 = 0.1
19 poptHG, pcovHG = curve_fit(HG, cos_theta, P_rayleigh, p0=[g0])
20 g_HG = poptHG[0]
21 print("HG fitting: g={:.6f} $\pm$ {:.6f}".format(*args: g_HG, np.sqrt(pcovHG[0][0])))
22 poptTTHG, pcovTTHG = curve_fit(TTHG, cos_theta, P_rayleigh, p0=[g0])
23 g_TTHG = poptTTHG[0]
24 print("TTHG fitting: g={:.6f} $\pm$ {:.6f}".format(*args: g_TTHG, np.sqrt(pcovTTHG[0][0])))
25 P_HG = HG(cos_theta, g_HG)
26 P_TTHG = TTHG(cos_theta, g_TTHG)
27 fig, ax = plt.subplots(figsize=(12,12),constrained_layout=True,subplot_kw={'projection': 'polar'})
28 ax.plot(*args: theta,P_rayleigh, c='black',ls='-',marker='.',label='Rayleigh Scattering Phase Function')
29 ax.plot(*args: theta,P_HG, c='red',ls='--',marker='s',label='HG Fitting: '+\
30         "g={:.6f} $\pm$ {:.6f}".format(*args: g_HG,np.sqrt(pcovHG[0][0])))
31 ax.plot(*args: theta,P_TTHG, c='lime',ls='--',marker='o',label='TTHG Fitting: '+\
32         "g={:.6f} $\pm$ {:.6f}".format(*args: g_TTHG,np.sqrt(pcovTTHG[0][0])))
33 ax.legend(loc='upper center',fontsize='xx-large')
34 ax.tick_params(which='major',labelsize='x-large')
35 plt.show()
36 fig.savefig(fname='Henyey-Greenstein.png', format='png', dpi=600, bbox_inches='tight', pad_inches=0.2)

```

